

Capstone Title: BALANGKAS : A GAMIFIED E-LEARNING AND ASSESSMENT SYSTEM FOR DATA STRUCTURES AND ALGORITHMS

Thank you for participating in this survey. This study aims to identify the specific academic and technical roadblocks students face when learning Data Structures and Algorithms (DSA). Your honest feedback will help in developing more effective learning strategies and support systems for IT professionals and students alike.

All responses are anonymous and will be used strictly for research purposes. Please rate the following statements based on your personal experience.

Name (Optional) _____

Year Level: ☐ 1st Year ☐ 2nd Year ☐ 3rd Year ☐ 4th Year
Status: ☐ Regular Student ☐ Irregular Student
Program: ☐ BSIT ☐ BSCS

A. Self-Assessment of Proficiency

1. What is your level of awareness with the concepts of Data Structure and Algorithm?

1. Novice / No Experience <i>prior knowledge or exposure to Data Structures and Algorithms (DSA).</i>	2. Basic Awareness <i>Basic awareness of DSA concepts, but limited understanding or experience.</i>	3. Working Knowledge <i>Understands common DSA concepts and can apply them with some guidance.</i>	4. Proficient <i>Strong understanding of No DSA concepts and can confidently apply them to solve problems independently.</i>
☐	☐	☐	☐

2. How long have you been studying Data Structures and Algorithms (DSA)?

- ☐ Less than a year
☐ 1 year
☐ More than 1 year
☐ Haven't studied at all

3. How difficult is it for you to understand DSA?

- ☐ 1. Very Easy ☐ 2. Easy ☐ 3. Moderate ☐ 4. Difficult ☐ 5. Very Difficult

4. On average, how much time do you spend mastering a specific concept of DSA until you can implement it independently?

- ☐ Less than 1 hour ☐ 1 – 3 hours ☐ A Full Day ☐ Multiple days / Ongoing study

B. Perceived Difficulty and Learning Obstacles

1. Provide the difficulty level of each DSA concepts using the scales:

1. Very Easy 2. Easy 3. Moderate 4. Difficult 5. Very Difficult

DSA Concept	Scale	DSA Concept	Scale
1. Arrays		9. Graphs	
2. Linked List		10. Iteration	
3. Stacks		11. Recursion	
4. Queues		12. Algorithms	
5. Trees			
6. Heaps			
7. Sets			
8. Maps			

2. What primary roadblock/s do you face when working and understanding Data Structures and Algorithms? Select All that applies.

- ☐ Understanding Concepts
- ☐ Applying Concepts
- ☐ Visualizing Data Structures
- ☐ Remembering Processes
- ☐ Implementing the Logic
- ☐ Others: _____

3. Please read each statement carefully and select the response that best reflects your personal experience. There are no right or wrong answers; your honest feedback is the most valuable asset to this study.

Scale: [1] Never | [2] Rarely | [3] Sometimes | [4] Often | [5] Always

	1	2	3	4	5
A. Study Habits & Engagement					
1. I dedicate time to review or practice DSA concepts outside of my required class hours.					
2. I use external resources (e.g., YouTube, AI tools, or textbooks) to supplement my understanding of DSA.					
3. I perform "dry-runs" or manual traces of data structures on paper to understand their behavior.					
B. Technical & Conceptual Roadblocks					

4. I find it difficult to mentally visualize how data moves within a structure (e.g., pointers, recursion, or tree branching).					
5. I understand the theoretical logic of a structure but struggle to implement it in actual code.					
6. I struggle to remember the specific steps or patterns of an algorithm shortly after learning them.					
7. I find it challenging to calculate and solve problems in DSA.					
8. I experience "total blockage" where I cannot proceed with a problem even with the help of hints or notes.					
C. Problem-Solving & Application					
9. I am confident in my ability to choose the most efficient data structure for a new, unseen problem.					
10. I find myself relying on external solutions (e.g., Google or AI) rather than solving the logic independently.					
11. I struggle to identify which specific algorithm (e.g., sorting vs. searching) is appropriate for a given task.					
12. I find the mathematical foundations of DSA more challenging than the actual coding implementation.					

4. Do you have other platforms used when learning DSA?
☐ No ☐ Yes, Specify _____
5. Would gamification (points, badges, levels) motivate you to learn DSA?
☐ Yes ☐ No ☐ Maybe
6. Which gamification features would you prefer? *Select all that applies*
- ☐ Points System
 - ☐ Badges/achievements
 - ☐ Leaderboards
 - ☐ Mini games
 - ☐ Progress Tracking
 - ☐ Reward System
 - ☐ Others: _____
7. If given a chance, what will be your perception of a gamified DSA learning platform? *Check all the applies.*
- ☐ I believe game elements (points, badges, leaderboards) would make me want to practice more consistently.
 - ☐ I think the interactive nature would help me "see" and understand complex structures (like Trees and Graphs) better than a textbook.
 - ☐ I feel that a game-like environment would make the difficult concepts of DSA feel less intimidating and more approachable.

- ☐ I believe seeing my progress compared to my peers would push me to master topics faster.
- ☐ I think a platform that lets me "level up" through coding challenges would better prepare me for technical interviews.
- ☐ I am concerned that the game elements might distract me from actually learning the deep logic and math required.
- ☐ I believe my learning style is better suited for traditional methods (lectures and labs) rather than games.